MA341
Applied ODE's
Handout 7

Definition 1. Matrix-vector product is defined as following: [1]

$$A\mathbf{v} = \underbrace{\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}}_{=A} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}}_{\mathbf{v}} = v_1 \underbrace{\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{bmatrix}}_{=a_{\downarrow 1}} + v_2 \underbrace{\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{bmatrix}}_{=a_{\downarrow 2}} + \dots + v_n \underbrace{\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{bmatrix}}_{=a_{\downarrow n}} \underset{also}{=} \begin{bmatrix} a_{1\rightarrow} \cdot \mathbf{v} \\ a_{2\rightarrow} \cdot \mathbf{v} \\ \vdots \\ a_{n\rightarrow} \cdot \mathbf{v} \end{bmatrix}$$

The last equality is a column of dot product (like in calc III) of rows of A with $\mathbf{v}$.

Definition 2. An n n-dimensional vectors v_j are linearly independent if and only if $det \begin{bmatrix} v_{\downarrow 1} & v_{\downarrow 2} & \dots & v_{\downarrow n} \end{bmatrix} \neq 0.$**??**^{0}

Definition 3. (**Higher order ODEs as a system of ODEs:**) Consider

$$y^{(n)} = a_{n-1}y^{(n-1)} + a_{n-2}y^{(n-2)} + \dots + a_0 y + f(t)$$

Let

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} = \begin{bmatrix} y(t) \\ y'(t) \\ \vdots \\ y^{(n-1)}(t) \end{bmatrix}$$

then

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}' = \mathbf{x}(t) = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & & \\ a_0 & a_1 & a_2 & \cdots & a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ f(t) \end{bmatrix}.$$

vfill

[1] For more about matrices - read Intro to Matrix Algebra on Moodle.

Definition 4. (**Euler's Method for Linear System of ODEs:**) Consider system of n ODEs with n unknown functions $x_j(t)$:[2]

$$\begin{aligned}
x_1'(t) &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + f_1(t)\\
x_2'(t) &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + f_2(t)\\
&\vdots\\
x_n'(t) &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n + f_n(t),
\end{aligned}$$

where $a_{ij} \in \mathbb{R}$. One write it in a matrix form

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{f}(t), \tag{1}$$

where A is the matrix of the coefficients of the system above, often denoted as $[A]_{ij} = a_{i,j}$.
Solution of the equation (**??**) is given by a sum of solution of the homogeneous equation ($\mathbf{f}(t) = 0$) and a private solution, as we used to have in second order ODEs, i. e. $\mathbf{x} = \mathbf{x}_h + \mathbf{x}_p$.
Homogeneous: Assuming $\mathbf{x}_h$ in the form of $\mathbf{x}(t) = ve^{rt}$ and substituting it into equation one get $rve^{rt} = Ae^{rt}$ or $Av = rv$, thus r is an eigenvalue of A and v its associated eigenvector. Hence, to find solution to y_h of (**??**) one need:

1. The eigenvalues - the n roots of characteristic polynomial $p(r) = det(A - rI)$, where I is the identity matrix. For example

$$\begin{aligned}
p(r) &= det\left(\begin{bmatrix} a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix} - r\begin{bmatrix} 1 & 0\\ 0 & 1\end{bmatrix}\right)\\
&= det\left(\begin{bmatrix} a_{11}-r & a_{12}\\ a_{21} & a_{22}-r\end{bmatrix}\right)\\
&= (a_{11}-r)(a_{22}-r) - a_{21}a_{12} = 0
\end{aligned}$$

2. The eigenvectors associated with the eigenvalues r_j's - normally, the solutions to $(A - r_jI)\mathbf{v_j} = 0$.

The solution to homogeneous equation is $\mathbf{x}_h = \sum_{j=1}^{n} c_jX_j(t)$, where c_j are the coefficients to be determined using IC's, and $X_j(t) \in S$ are the n linearly independent[3] fundamental solutions given as following:

- If $r_j \in \mathbb{R}$ solves $p(r)$, and $p(r) = (r - r_j)q(r)$, $q(r_j) \neq 0$, then $X_j(t) = e^{r_jt}v_j \in S$.

- If complex $r_j \in \mathbb{C}$ solves $p(r)$, (so also its complex conjugate $\bar{r}_j = r_{\tilde{j}}$) and $p(r) = (r - r_j)(r - \bar{r}_j)q(r)$, $q(r_j) \neq 0$. Instead of using complex valued

[2] In this course $n \leq 3$.
[3] When you solve system of ODEs - verify your S is actually a set of n linearly independent solutions! (how?)

functions $e^{r_j t}v_j, e^{\bar{r}_j t}\bar{v}_j$ ($v_{\bar{j}} = \bar{v}_j$), we (must!) use two real valued functions are given by $X_j(t) = \mathtt{Re}(e^{r_j t}v_j) \in S$ and $X_{\bar{j}}(t) = \mathtt{Im}(e^{r_j t}v_j) \in S$. Here

$$\mathtt{Re}(z) = \frac{1}{2}(z + \bar{z}) \underset{z=a+ib}{=} a \quad \text{and} \quad \mathtt{Im}(z) = \frac{1}{2}(z - \bar{z}) \underset{z=a+ib}{=} b.$$

- If eigenvalue r_j has multiplicity $k > 1$, i. e. $p(r) = (r - r_j)^k q(r)$, $q(r_j) \neq 0$, there are two options

  i. r_j has exactly K linearly independent eigenvectors $v_{j,1}, v_{j,2}, \ldots, v_{j,n}$, and so $X_{j,k}(t) = e^{r_j t}v_{j,k} \in S$, $k = 1, 2, \ldots K$.

  ii. r_j has less than K linearly independent eigenvectors. In this case we construct k linearly independent solutions $X_{j,k}(t) \in S$ as following.

$$\begin{aligned}
X_{j,1} &= v_{j,1}e^{r_j t} \\
X_{j,2} &= (v_{j,1}t + v_{j,2})e^{r_j t} \\
&\vdots \\
X_{j,1} &= (v_{j,1}\frac{t^{k-1}}{(k-1)!} + v_{j,2}\frac{t^{k-2}}{(k-2)!} + \cdots + v_{j,K})e^{r_j t}
\end{aligned}$$

  where the eigenvectors $v_{j,k}$ are obtained by solving:

$$\begin{aligned}
&(A - rI)v_{j,1} = 0 \\
&(A - rI)v_{j,k} = v_{j,k}, \quad k = 2, 3, \ldots, K.
\end{aligned}$$

Private: Let $X = \begin{bmatrix} X_{\downarrow 1}(t) & X_{\downarrow 2}(t) & \ldots & X_{\downarrow n}(t) \end{bmatrix}$. be a fundamental matrix, which columns $X_j(t) \in S$ are fundamental solutions. One writes $\mathbf{x}_h = X(t)\mathbf{c}$, where $\mathbf{c}$ is a vector of the coefficients c_j from above. With the matrix form of variation of parameters we assume $\mathbf{x}_p = X(t)\mathbf{u}(t)$. Substituting it into equation leads to $X(t)\mathbf{u}'(t) = \mathbf{f}(t)$. Thus, denote $\mathbf{y} = \mathbf{u}'$ and solve linear system of equations $X(t)\mathbf{y}(t) = \mathbf{f}(t)$, for example $\mathbf{y}(t) = X^{-1}(t)\mathbf{f}(t)$.**??**[0] Finally,

$$\mathbf{x}_p = X \int \mathbf{y}(t)dt = X \begin{bmatrix} \int y_1(t)dt \\ \int y_2(t)dt \\ \vdots \\ \int y_n(t)dt \end{bmatrix}$$